

SPECIFICATIONS FOR PROJECT REPORT

1. Report shall be computer typed (English- British, Font -Times Roman, Size-12 point) and printed on A4 size paper.
2. **The Report shall be spiral bound.** The name of the candidate, degree (specifying the branch), roll no, session, year of submission, name of the University including college name shall be printed in black on the cover
3. The report shall be typed on one side only with double space with a margin 3.5 cm on the left, 2.5 cm on the top, and 1.25 cm on the right and at bottom.
4. **Note: The number of pages in project final report must be not less than 20 pages.**
5. Conclusions must not exceed more than two pages.
6. References and Bibliography should be included in report.
7. The report must consist of following chapters
 - Chapter 1- Introduction
 - Chapter 2- Requirement Analysis and System Specification
 - Chapter 3- System Design
 - Chapter 4- Implementation and Testing
 - Chapter 5-Results and Discussions
 - Chapter 6-Conclusion and Future Scope
 - References
 - Appendix (if any)
 - Annexures-I,II,III
8. There should be only one copy per group of Final Report to be submitted in department.

SENTIMENTAL ANALYSIS OF TWITTER DATA USING BIG DATA ANALYTICS (24pt.)

PROJECT REPORT (14pt.)

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE OF (12pt.)

BACHELOR OF TECHNOLOGY
(Artificial Intelligence/ Machine Learning) (14pt.)



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Designation

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Abstract (Sample)

Packet Filtering firewalls can use a database of rules to decide which packets will be allowed to move in and out and from one network onto another. However with the increase in size of rule list, it's very hard to manage and validate the rules, which can also increase the cost of rule lookup and that may add significantly to latency. Packet filtering is the one of the major contemporary firewall design techniques. Implementation of such packet filter using Binary Decision Diagram (BDD) gives more advantages in terms of memory usage and look up time. In the case of the list-based packet filter firewall where rules are checked one by one for each incoming packet, the time taken to decide on a packet is proportional to the number of rules. The performance is improved with rule promotion but that is a slow and static kind of firewall implementation. In this work a BDD-based approach is presented which gives much better result in terms of number of comparisons or accesses the rule list make. This work presents the study, design and implementation of a packet filter firewall using binary decision diagram which provides faster processing of packets while maintaining the integrity of the original security policy. Results on large number of packets show that for most-accept packets, and for most-reject packets there is manifold reduction in such comparisons when BDD-based approach is used over list-based with promotion approach. The overall performance of a firewall is crucial in enforcing and administrating security, especially when the network is under attack. The continuous growth of the Internet, coupled with the increasing sophistication of the attacks, is placing stringent demands on firewall performance. In this work, a traffic-aware optimization framework is described to improve the operational cost of firewalls. Based on this framework a set of tools are designed that inspect and analyze both multidimensional firewall rules and traffic logs and construct the optimal equivalent firewall rules based on the observed traffic characteristics. The current work is the first to use traffic characteristics in firewall optimization. To evaluate the performance of current approach, a large set of firewall rules and traffic logs from a local LAN or at tens of enterprise networks managed by a Tier-1 service provider are evaluated. The evaluated results find these approaches very effective. In particular, current work has achieved more than 10 fold performance improvement by using the proposed traffic-aware firewall optimization.

SAMPLE SHEET-ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

We are highly grateful to the Dr. Surjit Kumar Gandhi, Principal, PCTE Institute of Engineering & Technology, Ludhiana, for providing this opportunity to carry out the project work at_____.

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Priyanka Walia

Deepak Kumar

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Note: The report of respective project should be as per prescribed format and in the same order though if some of the points are not applicable in regard with the concerned project, they might be omitted.